

What Can Be Certified in a Room-Guided Door Graph?

Research phase: PAUSE after 1 round, 0 ITERATE, 0 REVISE. Edit phase: readable note

Abstract

This note connects a survey of learning-augmented algorithms with a robot that builds a scene graph from room and door concepts. The thread asked whether the robot's claims about which rooms connect could be checked when its room model also guides what it observes. The peers proved stability for one door-detection threshold under a stated wall-error bound, and a sensing lower bound in a separate, simplified audit model. Neither result certifies the robot's complete graph or bounds its physical sensing cost. The verifier paused the thread because a usable claim needs a LiDAR visibility and error model, followed by experiments that test whether wall regions can be resolved at reasonable cost.

1 Introduction

Q surveys algorithms that use fallible predictions with formal performance guarantees [6]. It explains why component guarantees need separate assumptions when predictions change later observations, instances or costs. Transferring an error bound needs a metric and a bound on sensitivity. Its charge for prediction calls does not price robot travel or scans.

P describes a robot that builds an indoor scene graph from room and door concepts, without a prior global metric map [5]. Once it accepts a room, its walls guide the door search. A detected door can prompt a visit and then a crossing. Simplified tests recover the correct room connections, but false rooms can be accepted and later revision is not implemented.

The connection pursued was the room-to-door dependency. Can a slightly wrong room support a door decision? Can an unseen wall support a claim that no door is there? These questions join Q's concerns about error and missing feedback to P's detector.

2 What was done

The peers first traced how P's estimated walls filter LiDAR points for door detection. They compared this hard filter with Q's conditions for transferring an error guarantee. A small wall change can flip a point near the threshold. Ada derived a stability condition for the first filter with a fixed scan.

Emmy then modelled hidden openings as wall regions checked one by one. Her coupling argument measured the cost of claiming exact topology even with a correct no-door prediction. Ada checked it and noted that a real LiDAR view can cover several regions.

The peers finally compared evidence in P's graph. A door becomes nominal after a visit, before a crossing. A crossing witnesses a passage, but naming its rooms needs correct recognition and association. P also describes geometric loop closure through an uncrossed door. The peers proposed recording candidates, matches, crossings and unresolved regions separately. P does not supply this certificate.

3 Results and their scope

Proposition 1. *Fix a finite scan S , an estimated wall set $\widehat{W}$ and a true wall set W^* in the same coordinates. Let τ be P's wall-distance threshold, d_H the Hausdorff distance between wall sets, and ϵ a certified upper bound on their error. Define the smallest observed threshold margin by*

$$m = \min_{p \in S} \left| \text{dist}(p, \widehat{W}) - \tau \right|.$$

If $d_H(W^, \widehat{W}) \leq \epsilon < m$, every point in S makes the same pass or fail decision at the first wall-distance filter under either wall set.*

Here $\text{dist}(p, W)$ is the distance from scan point p to wall set W . It changes by at most the Hausdorff distance when the wall set changes. Thus a margin larger than the certified error prevents a threshold crossing. Ada's derivation in `ada/margin-gated-scene-graphs.md` applies this to P's 0.15 m filter. At the threshold, tiny wall changes can reverse the decision. Hence this binary gate has no global finite Lipschitz bound under wall distance, though Q allows other error-dependent guarantees.

Proposition 2. *In an audit-only model with n hidden wall regions, suppose each region has either a door or no door. Checking one region costs one unit and reveals only that region's state. Let δ , with $0 \leq \delta < 1/2$, be the permitted failure probability. If an algorithm must report every region correctly with probability at least $1 - \delta$ for every possible state under the same prediction, then, on the state with no doors, its expected number of distinct checks is at least $n(1 - 2\delta)$.*

For any region j , compare a no-door state with a state whose only door is in j . Give the algorithm the same prediction and random choices in both runs. Until it checks j , the runs look identical. If it omits j and correctly reports no doors in the first run, it must be wrong in the second. The allowed failure probability in each run therefore bounds the chance of omitting j by 2δ ; summing over regions gives the result. Emmy's proof is in `emmy/verification_cost.md`, and Ada checked it independently. The bound concerns independent unit-cost checks, not P's LiDAR views, travel, or sensing policy.

The results address different gaps. A margin supports a decision on measured points; an absence claim needs coverage and a sensor model. Neither proves P's graph correct. A crossing witnesses passage, while a named room pair also needs reliable identities and association.

4 Objections and unresolved questions

P's other door tests include derivative peaks, door widths, candidate pairing and association. The first-filter margin gives no bound for these decisions or for asynchronous graph updates.

P does not provide the deterministic wall-error bound assumed by the proposition. Its correct topology in simplified tests is evidence for those tests, not a general certificate. Likewise, the audit bound cannot be read as a cost bound for a LiDAR robot: one view may check many regions, and Q's charge for predictor calls does not price travel or sensing.

An unresolved label would avoid false claims if applied to every edge, but it would tell the robot little. A useful result needs a stated resolution target and an observation-cost measure. It also needs a rule for revising door claims when an accepted room changes. P proposes later fitness monitoring and alternative hypotheses, but does not implement that revision in the reported system [5].

The peers made a bounded check of related abstracts. *Active Semantic Perception* and *SCOUT* already connect uncertain scene graphs with active traversal [1, 4]. *Estimating Map Completeness in Robot Exploration* addresses a learned estimate of remaining map coverage, while *Landmark-based Distributed Topological Mapping and Navigation* uses designed landmark coverage conditions [2, 3]. Those checks show adjacent work; they do not establish novelty or exhaust the literature.

5 Why the thread stopped

The verifier accepted the two elementary arguments and their limits, but paused because neither yields a connectivity certificate or sensing-cost bound for P's robot. The smallest next step is to specify visibility and wall-error assumptions for one room, then test P's room-and-door scenario with wall perturbations near the threshold and occluded openings. The test should record asserted, absent and unresolved edges, wrong room-pair connections, and the sensing and travel needed to resolve wall regions. A useful certificate for that restricted setting would restart the larger question; the present record does not establish one.

References

- [1] Active Semantic Perception. URL: <https://arxiv.org/abs/2510.05430>.
- [2] Estimating Map Completeness in Robot Exploration. URL: <https://arxiv.org/abs/2406.13482>.
- [3] Landmark-based Distributed Topological Mapping and Navigation. URL: <https://arxiv.org/abs/2103.03741>.
- [4] SCOUT. URL: <https://arxiv.org/abs/2606.06721>.
- [5] Noé Zapata, Gerardo Pérez, Alejandro Torrejón, Pedro Núñez, and Pablo Bustos. Concept-Guided Exploration: Building Persistent, Actionable Scene Graphs, 2026. URL: <https://arxiv.org/abs/2608.23650>, [arXiv:2608.23650](https://arxiv.org/abs/2608.23650).
- [6] Hailiang Zhao, Xueyan Tang, Peng Chen, Zenong Ye, Tianjun Zhu, Jianwei Yin, and Shuiguang Deng. Learning-Augmented Algorithms: Guarantees, Construction Mechanisms, and System-Level Implications, 2026. URL: <https://arxiv.org/abs/2609.04787>, [arXiv:2609.04787](https://arxiv.org/abs/2609.04787).